

ARIAN RAHMAN ADITTA

Electronics & Communication Engineering Graduate | VLSI/EDA • Physical Design • ML for EDA

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[Academic Website](#) | [LinkedIn](#) | [Google Scholar](#) | [GitHub](#)

RESEARCH INTERESTS

• VLSI & Electronic Design Automation • Physical Design Automation • ML/GNN for EDA • RTL-to-QoR Modeling

EDUCATION

B.Sc. Engineering in Electronics and Communication Engineering

[2022 - 2026]

Khulna University of Engineering and Technology, Khulna, Bangladesh

CGPA: 3.71/4.00 | Class Rank: 5/60 (Top 8.3%)

Academic Distinction: Dean's Awards, 3rd and 4th Year

Final four term GPAs: 3.89, 4.00, 4.00, 3.88

Selected Coursework: VLSI Design & Nanotechnology, Analog Electronics, Microwave Engineering, Digital Communications, Antenna Engineering, Signals & Systems, Digital Signal Processing, Microprocessors & Microcomputers, Digital Electronics & Logic Circuits, Machine Learning

RESEARCH EXPERIENCE

Learning-Guided Constraint-Rich VLSI Floorplanning (GNN-SABLE)

Independent Research | 2026–Present | First-Author Manuscript Submitted to ICEEICT 2027

- Developing graph-based learning methods to guide B*-tree floorplanning under connectivity, geometric, outline and placement constraints while keeping exact legality explicitly enforced.
- Investigating the neural-legality gap between learned global floorplan structure and exact local geometric feasibility through controlled learning-and-optimization studies.
- Studying encoder-versus-decoder design choices to determine which aspects of physical design benefit from learning and which should remain algorithmically enforced.

Leakage-Aware RTL-to-QoR Prediction for Early Area and Timing Estimation

Research Collaboration with Zakia Sultana, Ph.D. Candidate, MSCAD Laboratory, University of Arkansas | Ongoing

- Developing machine-learning methods for early mapped-area and timing estimation directly from RTL before full technology mapping and static timing analysis.
- Working on source provenance, duplicate/near-duplicate grouping, leakage-aware evaluation, RTL structural feature engineering and reproducible modeling pipelines.
- Investigating structural-complexity effects, explainability and coverage/extrapolation warnings so predictions can be interpreted with appropriate engineering caution.

Open-Source RFIC Design and Verification of a 28-GHz Two-Stage Class-A SiGe Power Amplifier

Undergraduate Thesis, Department of ECE, KUET | Supervisor: Aysha Tabassum | 2026

- Designed a two-stage 28-GHz SiGe PA using the IHP SG13G2 Open PDK with an open-source-compatible RFIC flow spanning schematic-, process-, and EM-aware realization.
- Used machine learning as a supporting design-space exploration tool while retaining circuit simulation and RF verification as the primary evidence; a first-author thesis-derived manuscript has been submitted to ICCIT 2026.

PUBLICATIONS & MANUSCRIPTS

Peer-Reviewed Publications

[1] A. R. Aditta, P. K. Das, U. R. Arshee, and A. B. M. Aowlad Hossain, "SenseStick: A Bangla-English Edge-AI Cane for Road-Safety and Biomedical Reliability Monitoring," *IEEE BECITHCON*, 2026. [First Author]

[2] P. K. Das, S. M. R. Islam, and A. R. Aditta, "LMatchNet-V: Physics-Seeded Learning for Vendor-Aware RF L-Section Matching with Adaptive Safe-Library Search," *IEEE MARSCON*, 2026.

[3] M. A. Jahin, S. Soudeep, **A. R. Aditta**, M. F. Mridha, N. Fahad, and M. J. Hossen, "Vision Transformers for End-to-End Quark-Gluon Jet Classification from Calorimeter Images," *Communications in Computer and Information Science*, vol. 2640, Springer, pp. 135–150, 2025. [DOI: 10.1007/978-981-95-0988-1_10](#)

Submitted Manuscripts

[4] **A. R. Aditta**, M. M. Zaman, M. S. U. Alam, and S. M. R. Islam, "Graph Neural Network Guidance for B*-Tree Annealing in Constraint-Rich Outline-Aware Floorplanning," submitted to the *7th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT 2027)*. **[First Author]**

[5] **A. R. Aditta** and A. Tabassum, "Progressive PDK-to-EM Realization of a 28 GHz Two-Stage Class-A SiGe Power Amplifier in an Open-Source RFIC Flow," submitted to the *29th International Conference on Computer and Information Technology (ICCIT 2026)*. **[First Author]**

TECHNICAL SKILLS

- **RTL, Synthesis & Timing:** Verilog, Yosys, OpenSTA, RTL Analysis, Synthesis Workflows, Static Timing Analysis, Pre-Mapping Feature Extraction
- **Physical Design & Optimization:** VLSI Floorplanning, B*-Tree Methods, Simulated Annealing, Constraint Handling, Legality Checking, HPWL Analysis
- **ML for EDA:** Python, PyTorch, scikit-learn, XGBoost, Graph Neural Networks, Graph/Hypergraph Modeling, Tree-Based Models, Explainability
- **Supporting Circuit / RF Tools:** Qucs-S, ngspice, Xyce, KLayout, openEMS, IHP SG13G2 Open PDK
- **Scientific Computing & Reproducibility:** NumPy, Pandas, Matplotlib, Git, LaTeX, Leakage-Aware Evaluation

HONORS & AWARDS

Dean's Award, 3rd & 4th Year, Khulna University of Engineering & Technology

- Awarded for outstanding academic performance in consecutive senior academic years.

Technical Scholarship, Khulna University of Engineering and Technology (KUET)

- Awarded for academic performance during the first three undergraduate years.

ACADEMIC & RESEARCH LEADERSHIP

President, KUET Research Society, ECE

[2025-2026]

- Organized a department-wide pre-mid thesis competition involving 3rd- and 4th-year ECE students to promote research presentation, peer learning and early thesis engagement.
- Coordinated online graduate-study and research guidance sessions with KUET alumni pursuing M.S./Ph.D. degrees abroad, helping current students understand research preparation, graduate applications, and academic career pathways.

General Secretary, KUET Debating Society

[2023-2026]

- Led organizational activities and competitive debating initiatives, developing extensive experience in communication, team leadership, public speaking, team coordination and event management.

RELEVANT PROFESSIONAL EXPERIENCE

Campus Ambassador, Ulkasemi

[Aug 2024 – Sep 2025]

- Supported KUET–Ulkasemi semiconductor-industry engagement and student outreach; helped organize a five-day career and management training program for final-year students.

Data Analyst Intern, WEGRO

[Mar 2025 – Jul 2025]

- Analyzed business data, created decision-oriented visualizations, and contributed to data-driven process improvement in an agri-fintech startup.

ACADEMIC REFERENCES

Dr. A. B. M. Aowlad Hossain

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Dr. Sheikh Md. Rabiul Islam

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